

calc

1 Inputs

Parameter	Value	Unit	Source
Positive electrode thickness	7.56e-05	m	param_dump.txt Chen2020 (baseline)
Negative electrode thickness	8.52e-05	m	param_dump.txt Chen2020
Separator thickness (override)	7e-06	m	cell/params_y4.json
Positive CC thickness (override)	6e-06	m	cell/params_y4.json
Negative CC thickness (override)	5e-06	m	cell/params_y4.json
Positive electrode porosity	0.335	-	param_dump.txt
Negative electrode porosity	0.25	-	param_dump.txt
Separator porosity	0.47	-	param_dump.txt
Pos AM volume fraction	0.665	-	param_dump.txt
Neg AM volume fraction	0.75	-	param_dump.txt
Positive electrode density	3262	kg/m3	param_dump.txt
Negative electrode density	1657	kg/m3	param_dump.txt
Al collector density	2700	kg/m3	param_dump.txt
Cu collector density	8960	kg/m3	param_dump.txt
Electrode area	0.1027	m2	r6_y4_energy_dfn.json:area_m2
Electrolyte conductivity (override)	5	S/m	cell/params_y4.json
Cation transference number (override)	0.6	-	cell/params_y4.json
Electrolyte diffusivity (override)	9e-10	m2/s	cell/params_y4.json
Heat transfer coefficient (override)	26	W/m2/K	cell/params_y4.json
Cooling surface area (override)	0.01062	m2	cell/params_y4.json
Particle radii (override)	3e-06	m (both)	cell/params_y4.json
Nominal capacity	5	Ah	param_dump.txt

2 CapacityEnergy

Quantity	Value	Unit	Formula / Source
1C discharge capacity	5.0442	Ah	r6_y4_1c_dfn.json:capacity_ah
Discharge energy	18.2855	Wh	r6_y4_energy_dfn.json:energy_wh (integral V*I dt)
Midpoint voltage	3.832	V	r6_y4_energy_dfn.json:midpoint_voltage_v
4C T_max	327.6	K	r6_y4_4c_dfn.json:T_max_K
4C min anode potential	0.02	V	r6_y4_4c_dfn.json:anode_potential_v

3 EnergyDensity

Quantity	Value	Unit	Formula / Source
Gravimetric ED	535.72	Wh/kg	E_Wh / mass_kg; electrolyte excluded (contract)
Threshold	500.94	Wh/kg	task text (entry-0 contract)
PASS?	YES	None	535.72 >= 500.94
Volumetric ED	995.79	Wh/L	r6_y4_energy_dfn.json
Cell mass	34.132	g	r6_y4_energy_dfn.json:mass_kg
Stack thickness	178.8	um	r6_y4_energy_dfn.json:thickness_m
Cell volume	18.363	cm3	r6_y4_energy_dfn.json:volume_m3

4 NP_Mass

Quantity	Value	Unit	Formula / Source
Pos areal capacity	0.33	mAh/cm2	$t \times (1-e) \times \rho \times 201 \text{ mAh/g}$ (NMC811, literature default, annotated)
Neg areal capacity	0.394	mAh/cm2	$t \times (1-e) \times \rho \times 372 \text{ mAh/g}$ (graphite, literature default, annotated)
N/P ratio	1.19	-	neg/pos areal capacity (parameter set lacks capacity-density keys)
Mass positive layer	16.842	g	layer_kg_m2 x area
Mass negative layer	10.874	g	layer_kg_m2 x area
Mass Al CC	1.664	g	layer_kg_m2 x area
Mass Cu CC	4.601	g	layer_kg_m2 x area
Mass separator	0.151	g	layer_kg_m2 x area
Total mass (contract)	34.132	g	sum = r6_y4_energy_dfn.json:mass_kg
Electrolyte mass (est.)	6.152	g	pore vol x 1.2 g/cm3 (lit. default)

5 Process

Parameter	Value	Unit	Formula / Source
Pos areal density	163.99	g/m2	mass / area (16.842 g / 0.1027 m2)
Neg areal density	105.88	g/m2	mass / area (10.874 g / 0.1027 m2)
Pos compaction density	2.169	g/cm3	$\rho \times (1-e) / 1000$
Neg compaction density	1.243	g/cm3	$\rho \times (1-e) / 1000$
Electrolyte fill amount	6.152	g	pore vol 5.126 cm3 x 1.2 g/cm3 (lit. default)
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	None	design recommended; production tuning required